

ROLL NO:250701573

EXPERIMENT NO: 4. EXPERIMENT TO UNDERSTAND THE DATA PREPROCESSING
IN THE DATASET

PROGRAM:

```
import numpy as np
```

```
import pandas as pd
```

```
df = pd.read_csv("Employee_Dataset.csv")
```

```
print("ORIGINAL DATASET")
```

```
print(df)
```

```
print("\nDATASET INFORMATION")
```

```
df.info()
```

```
print("\nDEPARTMENT MODE")
```

```
print(df.Department.mode())
```

```
df["Department"] = df["Department"].fillna(df["Department"].mode()[0])
```

```
df["Age"] = df["Age"].fillna(df["Age"].median())
```

```
df["Salary"] = df["Salary"].fillna(round(df["Salary"].mean()))
```

```
print("\nDATASET AFTER FILLING MISSING VALUES")
```

```
print(df)
```

```

print("\nDUMMY VALUES")

print(pd.get_dummies(df.Department))

updated_dataset = pd.concat(
    [pd.get_dummies(df.Department), df.iloc[:, [1, 2, 3]]],
    axis=1
)

print("\nUPDATED DATASET")

print(updated_dataset)

updated_dataset["Promoted"] = updated_dataset["Promoted"].replace(
    ["No", "Yes"], [0, 1]
)

print("\nFINAL DATASET")

print(updated_dataset)

```

OUTPUT:

ORIGINAL DATASET

	Department	Age	Salary	Promoted
0	CSE	22.0	30000.0	No
1	IT	25.0	45000.0	Yes
2	ECE	24.0	38000.0	No
3	IT	NaN	42000.0	Yes
4	CSE	23.0	NaN	No
5	ECE	27.0	50000.0	Yes
6	IT	26.0	47000.0	No
7	CSE	21.0	32000.0	Yes
8	ECE	28.0	55000.0	No
9	CSE	24.0	36000.0	Yes

DATASET INFORMATION

<class 'pandas.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 4 columns):

#	Column	Non-Null Count	Dtype
0	Department	10 non-null	str
1	Age	9 non-null	float64
2	Salary	9 non-null	float64
3	Promoted	10 non-null	str

dtypes: float64(2), str(2)

memory usage: 452.0 bytes

DEPARTMENT MODE

0 CSE

Name: Department, dtype: str

DATASET AFTER FILLING MISSING VALUES

	Department	Age	Salary	Promoted
0	CSE	22.0	30000.0	No
1	IT	25.0	45000.0	Yes
2	ECE	24.0	38000.0	No
3	IT	24.0	42000.0	Yes
4	CSE	23.0	41667.0	No
5	ECE	27.0	50000.0	Yes
6	IT	26.0	47000.0	No
7	CSE	21.0	32000.0	Yes
8	ECE	28.0	55000.0	No
9	CSE	24.0	36000.0	Yes

DUMMY VALUES

	CSE	ECE	IT
0	True	False	False
1	False	False	True
2	False	True	False
3	False	False	True
4	True	False	False
5	False	True	False
6	False	False	True
7	True	False	False
8	False	True	False
9	True	False	False

UPDATED DATASET

	CSE	ECE	IT	Age	Salary	Promoted
0	True	False	False	22.0	30000.0	No
1	False	False	True	25.0	45000.0	Yes
2	False	True	False	24.0	38000.0	No

3	False	False	True	24.0	42000.0	Yes
4	True	False	False	23.0	41667.0	No
5	False	True	False	27.0	50000.0	Yes
6	False	False	True	26.0	47000.0	No
7	True	False	False	21.0	32000.0	Yes
8	False	True	False	28.0	55000.0	No
9	True	False	False	24.0	36000.0	Yes

FINAL DATASET

	CSE	ECE	IT	Age	Salary	Promoted
0	True	False	False	22.0	30000.0	0
1	False	False	True	25.0	45000.0	1
2	False	True	False	24.0	38000.0	0
3	False	False	True	24.0	42000.0	1
4	True	False	False	23.0	41667.0	0
5	False	True	False	27.0	50000.0	1
6	False	False	True	26.0	47000.0	0
7	True	False	False	21.0	32000.0	1
8	False	True	False	28.0	55000.0	0
9	True	False	False	24.0	36000.0	1